Electrical Energy Sources

Electrical energy is obtained from different sources of energy. These sources can be classified as :

1. Conventional (Non – Renewable) Sources
2. Non – Conventional (Renewable) Sources

1 Conventional (Non – Renewable) Sources

- These sources are available in limited quantity and will be exhausted in future. They cannot be renewed naturally in short time.

Examples :
Coal, Petroleum, Natural Gas, Nuclear fuel



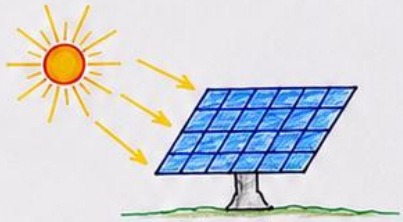
Coal Petroleum Natural Gas Nuclear Fuel

2 Non – Conventional (Renewable) Sources

- These sources are naturally replenished and are available in plenty. They are eco-friendly and will not be exhausted.

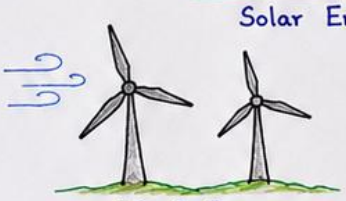
Examples :

(a) **Solar Energy :**
Energy obtained from the sun using solar cells or solar panels.



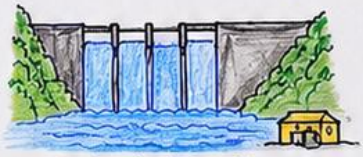
Solar Energy

(b) **Wind Energy :**
Energy obtained from the wind using wind turbines.



Wind Energy

(c) **Hydroelectric Energy :**
Energy obtained from the flow of water using turbines.



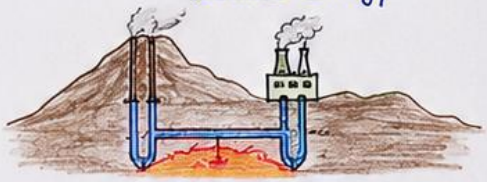
Hydroelectric Energy

(d) **Biomass Energy :**
Energy obtained from organic matter such as wood, crop residues, biogas, etc.



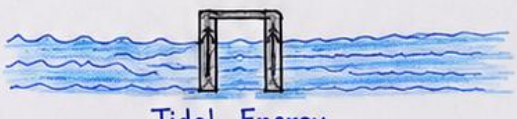
Biomass Energy

(e) **Geothermal Energy :**
Energy obtained from the heat inside the earth.



Geothermal Energy

(f) **Tidal Energy :**
Energy obtained from the rise and fall of sea water tides.



Tidal Energy

CIRCUIT CONCEPT

An electric circuit is a closed path in which electric current can flow. It consists of various electrical and electronic elements connected together by conductors.

1. Definition :

- A circuit is an interconnection of electrical elements forming a closed loop through which current flows from the source, through the elements and returns back to the source.

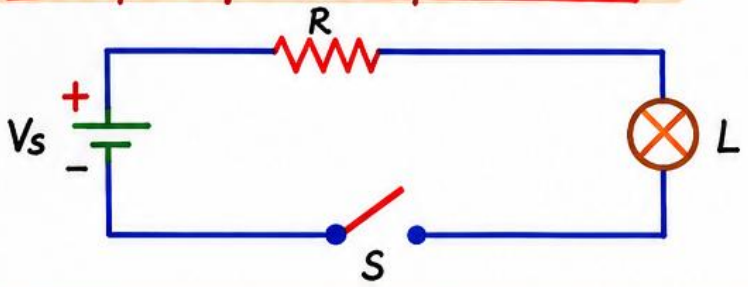
2. Basic Elements of a Circuit :

- (i) Source : Provides electrical energy.
- (ii) Conductor : Provides a path for current flow.
- (iii) Load : Utilizes electrical energy.
- (iv) Control device : Controls the flow of current.
- (v) Protective device : Protects the circuit from faults and damage.

3. Types of Circuits :

- (i) Open Circuit : The circuit in which the path is incomplete, hence no current flows.
- (ii) Closed Circuit : The circuit in which the path is complete, hence current flows.

4. Example of a Simple Circuit :



- V_s – Voltage Source
- R – Resistor
- L – Lamp (Load)
- S – Switch

5. Characteristics of an Ideal Circuit :

- The circuit elements are considered ideal.
- The conductors are assumed to have zero resistance.
- The sources are considered to be constant.
- There are no losses in the circuit.

Note : For current to flow in a circuit, there must be a complete closed path from the positive terminal of the source to the negative terminal.

Electric Element Resistance

1. Introduction :

Resistance is the property of an electric element or material to oppose the flow of electric current. It dissipates electrical energy in the form of heat.

The SI unit of resistance is Ohm (Ω).

2. Symbol :



The symbol of resistance is shown above.

3. Ohm's Law :

At constant temperature, the current (I) flowing through a resistance (R) is directly proportional to the potential difference (V) across it.

$$V \propto I \quad \text{or} \quad V = IR$$

Where,

V = Voltage across resistance (volt)

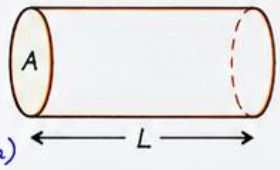
I = Current through resistance (ampere)

R = Resistance (ohm)

4. Resistance of a Conductor :

The resistance of a conductor depends on its material, length, cross-sectional area and temperature.

$$R = \rho \frac{L}{A}$$



Where,

ρ = Resistivity of the material ($\Omega \cdot m$)

L = Length of the conductor (m)

A = Cross-sectional area (m^2)

R = Resistance (Ω)

5. Factors Affecting Resistance :

- **Length (L)** : Resistance is directly proportional to the length of the conductor.

$$R \propto L$$

- **Area of cross-section (A)** : Resistance is inversely proportional to the area.

$$R \propto \frac{1}{A}$$

- **Nature of material (ρ)** : Resistance is directly proportional to the resistivity.

$$R \propto \rho$$

- **Temperature (T)** : For metallic conductors, resistance increases with temperature.

$$R_T = R_0 [1 + \alpha (T - T_0)]$$

Where,

R_T = Resistance at temperature T (Ω)

R_0 = Resistance at reference temperature T_0 (Ω)

α = Temperature coefficient of resistance (per °C)

T = Temperature (°C), T_0 = Reference temperature (°C)

6. Resistivity (ρ) of Some Materials

Material	Resistivity ρ ($\Omega \cdot m$)
Silver	1.59×10^{-8}
Copper	1.68×10^{-8}
Aluminium	2.82×10^{-8}
Iron	9.71×10^{-8}
Nichrome	1.10×10^{-6}
Manganin	4.40×10^{-7}

7. Applications :

* Used in electric heating devices.
* Current limiting in circuits.
* Potential divider and voltage divider networks.
* Protection circuits and load resistors.
* Used in measuring instruments.

Key Points :

* Ohm's law is valid for ohmic (linear) conductors.
* Resistance converts electrical energy into heat.
* Good conductors have low resistance and vice versa.

Electric Circuit Inductance

1. Introduction :

Inductance is the property of a conductor or a circuit by which a change in current flowing through it produces an induced emf in it or in a nearby conductor.

It opposes any change in current.

2. Inductance (L) :

Inductance (L) is the ratio of the flux linkage (λ) with a circuit to the current (I) producing it.

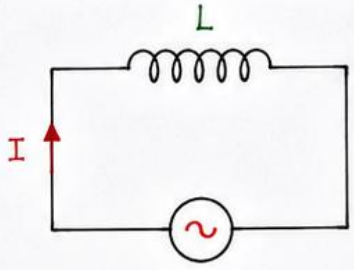
$$L = \frac{\lambda}{I} \quad \text{Henry (H)}$$

Where,

- λ = Flux linkage (weber-turn)
- I = Current (ampere)
- L = Inductance (henry)

3. Self Inductance :

The property of a coil to induce emf in itself due to change in current flowing through it is called self inductance.



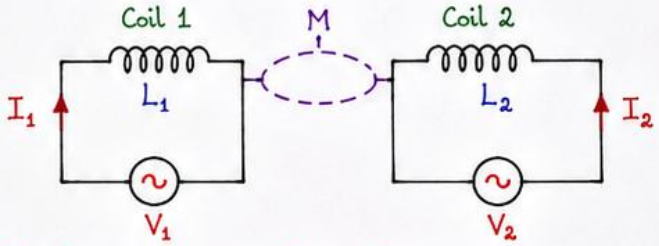
Induced emf,

$$E = -L \frac{dI}{dt}$$

Unit of Inductance : Henry (H)

4. Mutual Inductance :

When a change in current in one coil induces emf in another nearby coil, the property is called mutual inductance (M).



5. Energy Stored in Inductor :

The energy stored in an inductor when current I flows through it is

$$W = \frac{1}{2} L I^2 \quad \text{Joule}$$

6. Inductance of Some Common Coils :

- Long solenoid : $L = \frac{\mu N^2 A}{\ell}$ Henry
- Toroid : $L = \frac{\mu N^2 A}{2\pi r}$ Henry
- Air-core coil : $L \propto N^2$

N = Number of turns
A = Cross-sectional area (m^2)
ℓ = Length of coil (m)
r = Mean radius of toroid (m)
μ = Permeability of core material

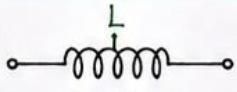
7. Effects of Inductance :

- Inductor opposes the change in current.
- At steady state (DC), inductor behaves as a short circuit.
- At instant switching (t = 0), inductor opposes the current.
- Inductance causes voltage to lead current by 90° in a pure inductive circuit.

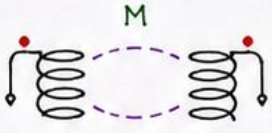
8. Applications :

- Filters in power supplies
- Chokes and reactors
- Energy storage
- Tuned circuits and oscillators
- Induction heating
- Transformers

Important Symbols :



Inductor



Mutually coupled inductors

λ = Flux linkage (weber-turn)
I = Current (ampere)
L = Inductance (henry)
M = Mutual inductance (henry)

Electric Circuit Capacitance

1. Introduction :

Capacitance is the ability of a circuit element or system to store electric charge when a potential difference is applied across it.

2. Definition :

$$\text{Capacitance (C)} = \frac{\text{Charge stored (Q)}}{\text{Potential difference (V)}}$$

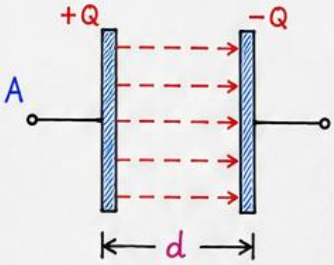
$$C = \frac{Q}{V}$$

SI Unit : Farad (F)

1 F = 1 Coulomb / Volt

3. Capacitor :

A capacitor is a passive two-terminal circuit element that stores energy in the electric field between two conducting plates separated by an insulating material (dielectric).



A = Area of plate (m^2)

d = Separation (m)

ε = Permittivity of dielectric (F/m)

$$C = \frac{\varepsilon A}{d}$$

where,

$\varepsilon = \varepsilon_0 \varepsilon_r$

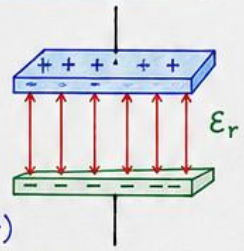
ε_0 = Permittivity of free space

$= 8.854 \times 10^{-12}$ F/m

ε_r = Relative permittivity (dielectric constant)

4. Capacitance of Parallel Plate Capacitor :

$$C = \frac{\varepsilon A}{d}$$



where,

$\varepsilon = \varepsilon_0 \varepsilon_r$

A = Area of each plate (m^2)

d = Separation between plates (m)

5. Energy Stored in a Capacitor :

$$W = \frac{1}{2} C V^2$$

where,

W = Energy stored (Joule)

C = Capacitance (F)

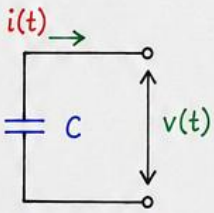
V = Potential difference (Volt)

6. Voltage – Current Relationship :

$$i(t) = C \frac{dv(t)}{dt}$$

or

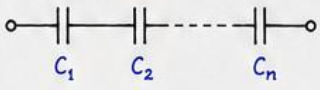
$$v(t) = \frac{1}{C} \int i(t)\, dt$$



This shows that current through a capacitor is proportional to the rate of change of voltage across it.

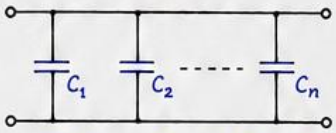
7. Combination of Capacitors :

(i) Series Combination :



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

(ii) Parallel Combination :

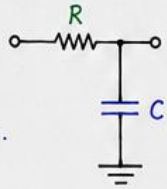


$$C_{eq} = C_1 + C_2 + \cdots + C_n$$

where, C_{eq} = Equivalent Capacitance

8. Applications :

- Used in filtering and timing circuits.
- Used in energy storage systems.
- Used in coupling and decoupling circuits.
- Used in power factor improvement.



9. Key Points :

- ★ Capacitor stores energy in the electric field.
- ★ It opposes sudden change in voltage.
- ★ In DC steady state, capacitor behaves as open circuit.
- ★ In AC circuits, capacitive reactance $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ (ohm)

10. Capacitive Reactance :

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} \ (\Omega)$$

where,

ω = Angular frequency (rad/s)

f = Frequency (Hz)

Capacitance is widely used in electronic circuits for filtering, timing, energy storage and signal processing.

VOLTAGE AND CURRENT SOURCES

① VOLTAGE SOURCE

A voltage source is an ideal circuit element that maintains a constant voltage across its terminals, irrespective of the current flowing through it.

Symbol :



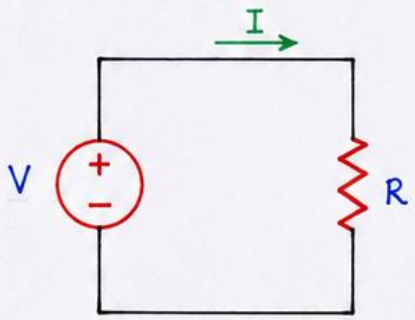
Equation : $V = \text{constant}$

Example : A 12 V battery supplies 12 volts across its terminals regardless of the current.

Characteristics :

- Internal resistance = 0
- Voltage across the source is fixed.
- Current can vary depending on the external circuit.

Practical representation :



V is constant but I depends on R.

---

② CURRENT SOURCE

A current source is an ideal circuit element that maintains a constant current through it irrespective of the voltage across its terminals.

Symbol :



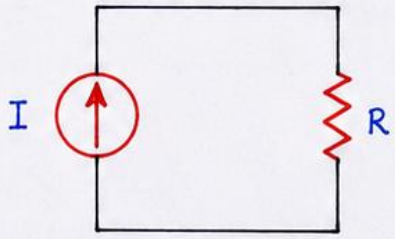
Equation : $I = \text{constant}$

Example : A current regulator in an LED driver supplies a constant current of 350 mA to the LED.

Characteristics :

- Internal resistance = ∞ (infinite)
- Current through the source is fixed.
- Voltage across the source can vary depending on the external circuit.

Practical representation :



I is constant but V depends on R.

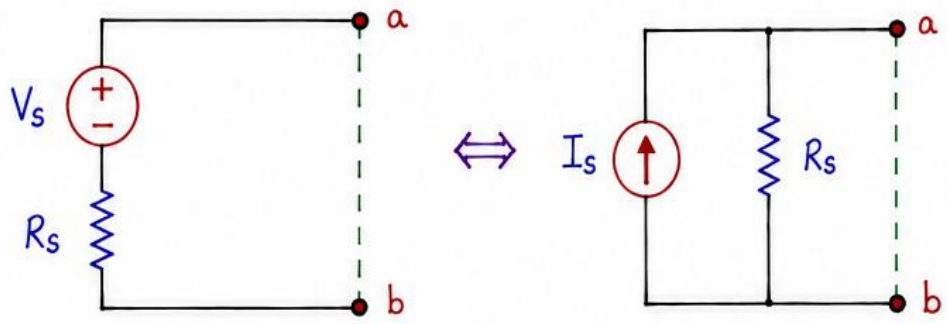
COMPARISON

Feature	Voltage Source	Current Source
Maintains	Constant voltage	Constant current
Can vary	Current can vary	Voltage can vary
Internal resistance	Zero (0)	Infinite (∞)
Ideal condition	I can be any value	V can be any value
Example	Battery	Current regulator

Source Transformation

Source transformation is the process of converting a practical voltage source into an equivalent practical current source and vice versa without changing the voltage and current relationship at the terminals.

1. Voltage Source to Current Source



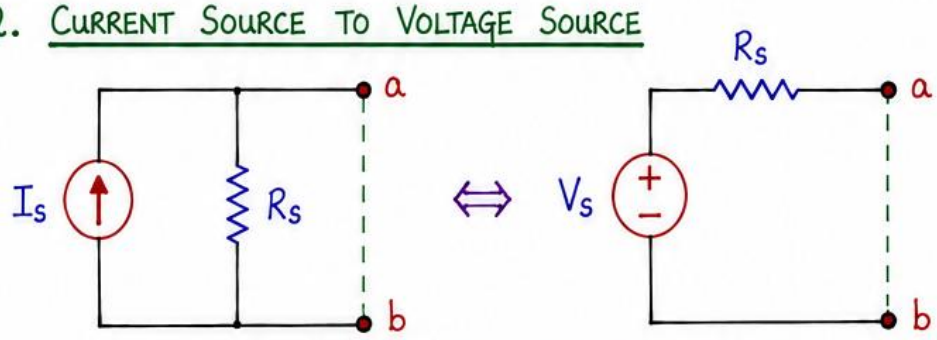
$$I_s = \frac{V_s}{R_s}$$

(I_s = Current source value)

R_s = Same resistance

The current source is in parallel with the same resistance.

2. Current Source to Voltage Source



$$V_s = I_s \times R_s$$

(V_s = Voltage source value)

R_s = Same resistance

The voltage source is in series with the same resistance.

Key Points:

* Both transformed circuits are electrically equivalent at terminals a-b.
* Applicable for any linear circuit.
* Power remains the same in both forms.

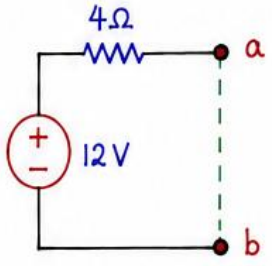
Applications:

- Simplification of circuit analysis.
- Thevenin and Norton equivalent circuits.
- Useful in network theorems.

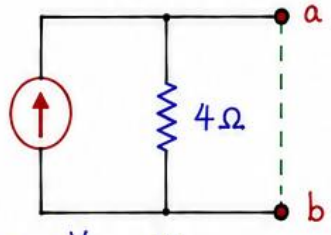
Example:

Given: $V_s = 12\,V$, $R_s = 4\,\Omega$

Voltage source circuit



Equivalent current source circuit



$$I_s = \frac{V_s}{R_s} = \frac{12}{4} = 3\,A$$

Note: Power remains the same in both forms. ($P = V \times I$)

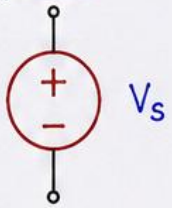
Electrical Independent Sources

An independent source is an ideal source that supplies a specified voltage or current regardless of any other circuit variables. The value of an independent source does not depend on any other voltage or current in the circuit.

1. Independent Voltage Source

An independent voltage source maintains a constant voltage across its terminals, irrespective of the current flowing through it or other circuit conditions.

Symbol:



Symbol of Independent Voltage Source

Characteristics:

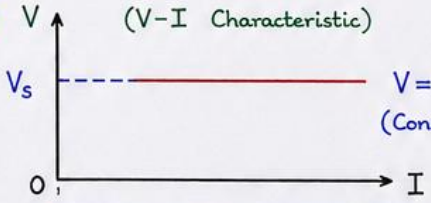
- Delivers a fixed voltage V_s.
- Voltage across the source can vary depending on the circuit.
- Internal resistance of an ideal voltage source is zero.

Example:

A 12 V battery idealized as an independent voltage source of $V_s = 12$ V.

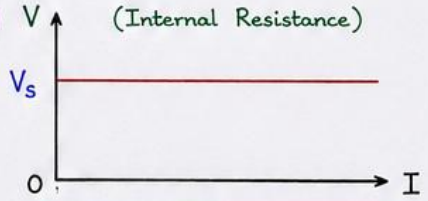
Characteristics Figures (Voltage Source)

① V (V-I Characteristic)



Voltage remains constant at V_s for any current I.

② V (Internal Resistance)

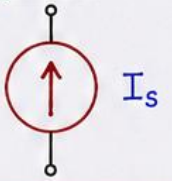


Ideal internal resistance $R_{in} = 0\ \Omega$.

2. Independent Current Source

An independent current source maintains a constant current through its terminals, irrespective of the voltage across it or other circuit conditions.

Symbol:



Symbol of Independent Current Source

Characteristics:

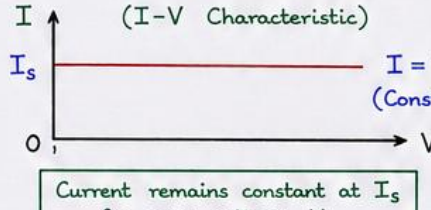
- Delivers a fixed current I_s.
- Voltage across the source can vary depending on the circuit.
- Internal resistance of an ideal current source is infinite.

Example:

A current source that delivers $I_s = 2$ A regardless of the voltage across it.

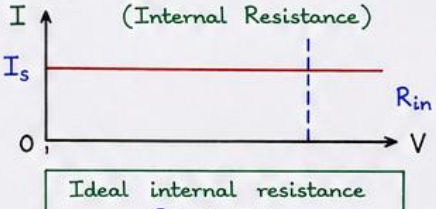
Characteristics Figures (Current Source)

① I (I-V Characteristic)



Current remains constant at I_s for any voltage V.

② I (Internal Resistance)



Ideal internal resistance $R_{in} = \infty$:

Key Points:

- Independent sources are idealized components.
- They are used to model real sources like batteries (voltage source) and current generators.
- They are not controlled by any other circuit variable.

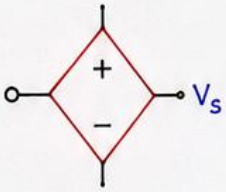
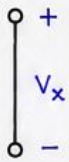
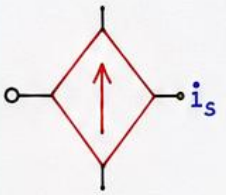
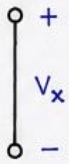
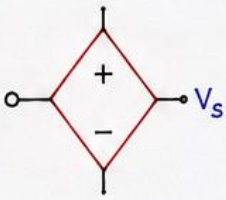
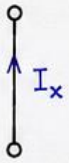
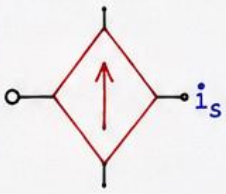
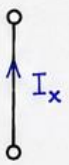
Electrical Dependent Sources

Dependent (controlled) sources are sources whose voltage or current depends on some other voltage or current elsewhere in the circuit.

Types of Dependent Sources :

There are four types of dependent (controlled) sources.

① Voltage Controlled Voltage Source (VCVS)
② Voltage Controlled Current Source (VCCS)
③ Current Controlled Voltage Source (CCVS)
④ Current Controlled Current Source (CCCS)

Type	Symbol	Controlling Variable	Equation	Unit of Gain	Characteristics
① Voltage Controlled Voltage Source (VCVS)			$V_s = \mu V_x$ μ = Voltage gain (dimensionless)	— (dimensionless)	• Output is a voltage. • Polarity of V_s is as shown. • μ can be >1, <1, =1 or negative.
② Voltage Controlled Current Source (VCCS)			$i_s = g V_x$ g = Transconductance (Siemens)	Siemens (S)	• Output is a current. • Direction of i_s is as shown (arrow). • g can be positive or negative.
③ Current Controlled Voltage Source (CCVS)			$V_s = r i_x$ r = Transresistance (Ohms)	Ohms (Ω)	• Output is a voltage. • Polarity of V_s is as shown. • r can be positive or negative.
④ Current Controlled Current Source (CCCS)			$i_s = \beta i_x$ β = Current gain (dimensionless)	— (dimensionless)	• Output is a current. • Direction of i_s is as shown (arrow). • β can be >1, <1, =1 or negative.

Characteristics :

- Dependent sources do not supply energy on their own; they control or transfer energy.
- Their value is determined by another voltage or current in the circuit.
- They can have positive, negative or zero gain.
- They are widely used in modeling amplifiers, transistors, op-amps and active circuits.
- In circuit analysis, dependent sources are treated like independent sources.

Key Points :

- The diamond shape indicates a dependent source.
- The arrow indicates a current source.
- The + and − signs indicate the reference polarity of voltage.
- The controlling variable (V_x or i_x) must be measured in the same circuit.
- Units of gain are: V/V, A/V, V/A, A/A (depending on source type).

Kirchhoff's Laws

Kirchhoff's laws are fundamental Laws used in the analysis of electrical circuits. They are based on the conservation of charge and energy.

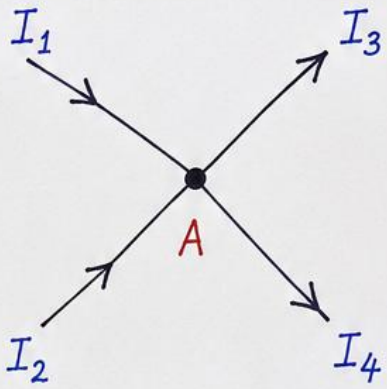
1. Kirchhoff's Current Law (KCL)

$\Sigma I = 0$

(At any node)

→ It is based on the conservation of charge.

→ It states that the algebraic sum of all currents at any node (junction) in a circuit is zero.



Mathematically, $\boxed{I_1 + I_2 + I_3 + \ldots + I_n = 0}$

Example: In the given node,

$$I_1 + I_2 = I_3 + I_4$$

or, $I_1 + I_2 - I_3 - I_4 = 0$

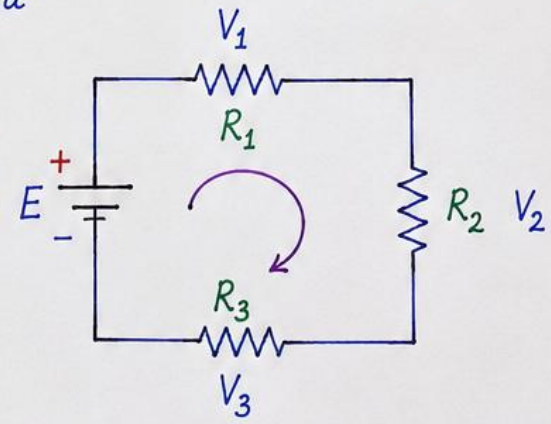
2. Kirchhoff's Voltage Law (KVL)

$\Sigma V = 0$

(Around any closed loop)

→ It is based on the conservation of energy.

→ It states that the algebraic sum of all voltages around any closed loop in a circuit is zero.



Mathematically, $\boxed{V_1 + V_2 + V_3 + \ldots + V_n = 0}$

Example: In the given loop,

$$+ V_1 - V_2 - V_3 = 0$$

Note:
- KCL is applied at nodes (junctions).
- KVL is applied in closed loops (meshes).

NETWORK REDUCTION TECHNIQUES

Network reduction is the process of simplifying a complex network into an equivalent circuit using basic transformations.

The commonly used techniques are :

1. Series Reduction
2. Parallel Reduction
3. Series – Parallel Reduction
4. Star / Delta Transformations

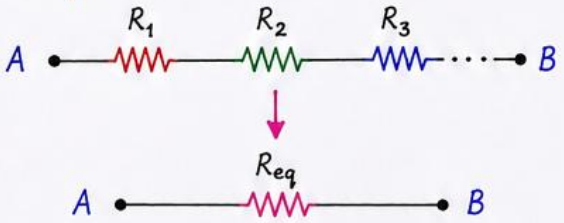
1. SERIES REDUCTION :

Elements are connected end to end so that the same current flows through them.

For resistors in series,

$$R_{eq} = R_1 + R_2 + R_3 + \cdots + R_n$$

Example :



$$R_{eq} = R_1 + R_2 + R_3 + \cdots + R_n$$

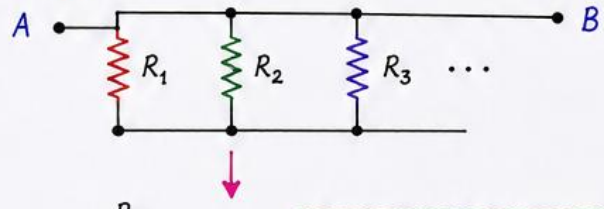
2. PARALLEL REDUCTION :

Elements are connected across the same two nodes so that the same voltage appears across them.

For resistors in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$$

Example :



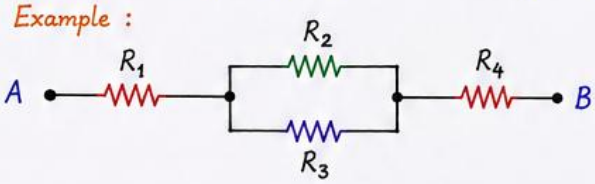
$$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}\right)^{-1}$$

3. SERIES – PARALLEL REDUCTION :

A network may contain series and parallel combinations.

We reduce step by step until a single equivalent resistance remains.

Example :



Step 1 : Reduce the parallel Bart $(R_2 \,//\, R_3)$

$$R_{23} = \frac{R_2 R_3}{R_2 + R_3}$$

Step 2 : Now the circuit becomes series.

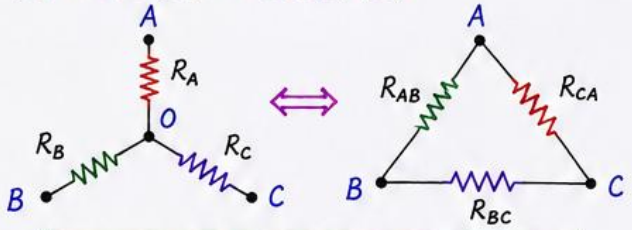
$$R_{eq} = R_1 + R_{23} + R_4$$

$$R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3} + R_4$$

4. STAR / DELTA TRANSFORMATIONS :

These transformations are used to convert a star (Y) network into an equivalent delta (Δ) network and vice versa.

(a) STAR (Y) → DELTA (Δ)

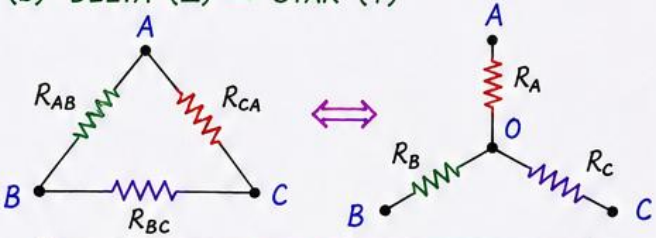


$$R_{AB} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C}$$

$$R_{BC} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_A}$$

$$R_{CA} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B}$$

(b) DELTA (Δ) → STAR (Y)



$$R_A = \frac{R_{AB} R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_B = \frac{R_{AB} R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_C = \frac{R_{BC} R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

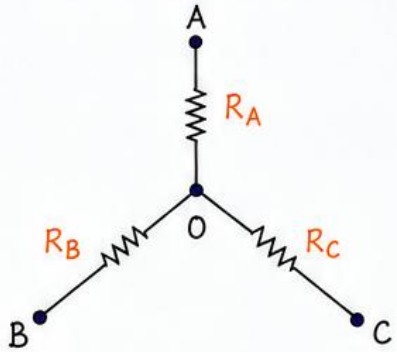
★ These techniques simplify complex circuits and make analysis faster and easier.

STAR TO DELTA TRANSFORMATION AND DERIVATION

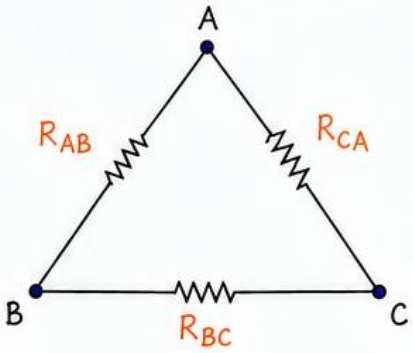
1) **Star (Y) Network** : It is a 3 terminal network in which three resistances are connected to a common point.

2) **Delta (Δ) Network** : It is a 3 terminal network in which three resistances are connected in closed loop.

Star (Y) Network



Delta (Δ) Network

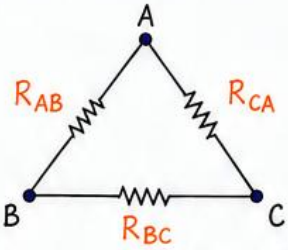


Derivation : For equivalence, the resistance (impedance) between any two terminals must be same in both networks.

1) Resistance between A and B

In Star (Y) network,

$$R_{AB}(Y) = R_A + R_B \quad \text{—①}$$



In Delta (Δ) network, Between A and B, the path is direct R_{AB} in parallel with $(R_{CA} + R_{BC})$.

$$R_{AB}(\Delta) = R_{AB} \parallel (R_{CA} + R_{BC}) = \frac{R_{AB}(R_{CA} + R_{BC})}{R_{AB} + R_{CA} + R_{BC}} \quad \text{—②}$$

Equating ① and ②

$$R_A + R_B = \frac{R_{AB}(R_{CA} + R_{BC})}{R_{AB} + R_{CA} + R_{BC}}$$

$$\Rightarrow \boxed{R_{AB} = \frac{(R_A + R_B)(R_A + R_C + R_B)}{R_C}} \quad \text{— (i)}$$

2) Resistance between B and C

In Star (Y) network,

$$R_{BC}(Y) = R_B + R_C \quad \text{—③}$$

In Delta (Δ) network,

$$R_{BC}(\Delta) = R_{BC} \parallel (R_{AB} + R_{CA}) = \frac{R_{BC}(R_{AB} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{—④}$$

Equating ③ and ④

$$\Rightarrow \boxed{R_{BC} = \frac{(R_B + R_C)(R_A + R_B + R_C)}{R_A}} \quad \text{— (iii)}$$

3) Resistance between C and A

In Star (Y) network,

$$R_{CA}(Y) = R_C + R_A \quad \text{—⑤}$$

In Delta (Δ) network,

$$R_{CA}(\Delta) = R_{CA} \parallel (R_{AB} + R_{BC}) = \frac{R_{CA}(R_{AB} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \quad \text{—⑥}$$

Equating ⑤ and ⑥

$$\Rightarrow \boxed{R_{CA} = \frac{(R_C + R_A)(R_A + R_B + R_C)}{R_B}} \quad \text{— (iiii)}$$

Hence ; Star (Y) to Delta (Δ) Transformation Formulae :

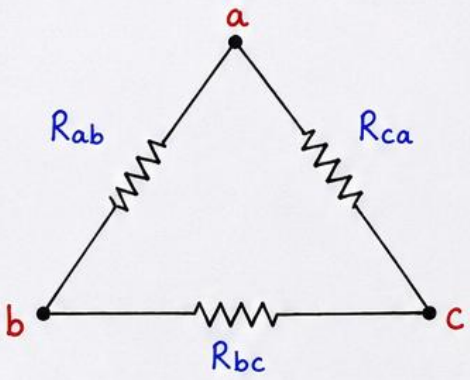
$$R_{AB} = \frac{(R_A + R_B)(R_A + R_B + R_C)}{R_C} \; ; \; R_{BC} = \frac{(R_B + R_C)(R_A + R_B + R_C)}{R_A} \; ; \; R_{CA} = \frac{(R_C + R_A)(R_A + R_B + R_C)}{R_B}$$

* These equations are used to convert Star (Y) network to Delta (Δ) network.

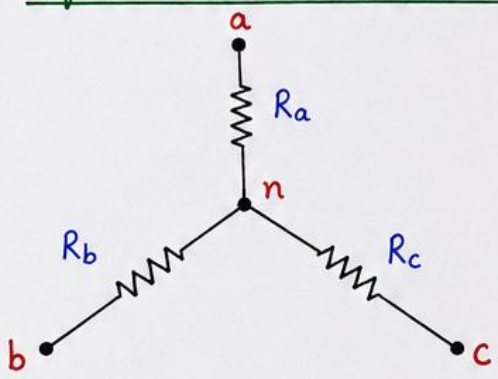
Delta To Star Transformation And Derivation

In a Δ (Delta) connected network with resistances R_{ab}, R_{bc}, R_{ca}, the equivalent Star (Y) network has resistances R_a, R_b, R_c connected to the same terminals a, b, c.

1. Δ (Delta) Network



2. Equivalent Star (Y) Network



3. Derivation :-

For equivalence, the resistance between any two terminals must be the same in both networks.

In Δ Network :-

$$R_{ab} = R_{ab}$$
$$R_{bc} = R_{bc}$$
$$R_{ca} = R_{ca}$$

In Y Network :-

$$R_{ab} = R_a + R_b$$
$$R_{bc} = R_b + R_c$$
$$R_{ca} = R_c + R_a$$

From above,

$$R_a + R_b = R_{ab} \quad — (1)$$
$$R_b + R_c = R_{bc} \quad — (2)$$
$$R_c + R_a = R_{ca} \quad — (3)$$

Adding (1), (2) and (3)

$$2(R_a + R_b + R_c) = R_{ab} + R_{bc} + R_{ca}$$

$$\therefore \quad R_a + R_b + R_c = \frac{R_{ab} + R_{bc} + R_{ca}}{2} \quad — (4)$$

From (1) and (4) :-

$$R_a = R_{ab} - R_b$$
$$= R_{ab} - \left(\frac{R_{al} + R_b + R_c - R_a}{2}\right)$$
$$= \frac{2R_{ab} - (R_b + R_c)}{2}$$
$$= \frac{R_{ab} - R_b - R_c}{2}$$

$$\boxed{R_a = \frac{R_{ab}\, R_{ca}}{R_{ab} + R_{bc} + R_{ca}}}$$

Similarly,

$$\boxed{R_b = \frac{R_{ab}\, R_{bc}}{R_{ab} + R_{bc} + R_{ca}}}$$

$$\boxed{R_c = \frac{R_{bc}\, R_{ca}}{R_{ab} + R_{bc} + R_{ca}}}$$

Result (Δ to Y Transformation) :-

$$R_a = \frac{R_{ab}\, R_{ca}}{R_{ab} + R_{bc} + R_{ca}}, \quad R_b = \frac{R_{ab}\, R_{bc}}{R_{ab} + R_{bc} + R_{ca}}, \quad R_c = \frac{R_{bc}\, R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

Superposition Theorem

Statement :-

In any linear, bilateral network containing two or more independent sources, the current through or voltage across any element is equal to the algebraic sum of the currents or voltages caused by each independent source acting alone, with all other independent sources replaced by their internal resistances.

Note :-

* Dependent sources are always active.
* Superposition theorem is not applicable to power.

Conditions :-

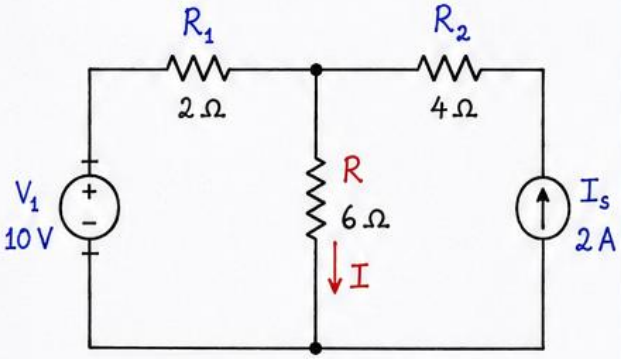
1. The network must be linear.
2. The network must be bilateral.
3. Only independent sources are to be considered.

Procedure :-

1. Consider one independent source at a time.
2. Replace all other independent sources by their internal resistances.
   - Ideal voltage source → replace by short circuit ($R = 0.\Omega$)
   - Ideal current source → replace by open circuit ($R \to \infty$)
3. Find the current or voltage in the element due to that source.
4. Repeat steps 1 to 3 for all independent sources.
5. The algebraic sum of all individual contributions gives the total current or voltage.

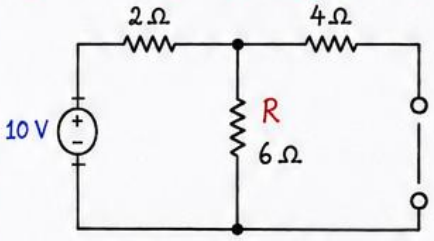
Example :- Find the current I through resistor R using superposition theorem.

Given circuit :-



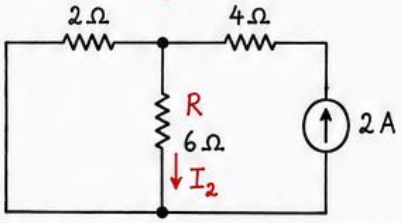
Solution :-

1) Due to V_1 only (I_s opened)



$$R_{eq} = 2 + \frac{4 \times 6}{4 + 6}$$
$$= 2 + \frac{24}{10} = 2 + 2.4$$
$$= 4.4\ \Omega$$
$$I_1 = \frac{10}{4.4} = 2.273\ A$$
(downward)

2) Due to I_s only (V_1 shorted)



Using current division,

$$I_2 = 2 \times \frac{2}{2 + 4}$$
$$= 2 \times \frac{2}{6} = \frac{2}{3}$$
$$= 0.667\ A$$
(downward)

Step 3 :- Total current through R,

$$I = I_1 + I_2 = 2.273 + 0.667 = 2.940\ A$$
(downward)

Conclusion :-

Current through 6Ω resistor

$I = 2.940$ A downward.

Superposition Theorem and Derivation

1. Superposition Theorem :

In any linear bilateral network containing two or more independent sources, the current through or voltage across any element is equal to the algebraic sum of the currents or voltages produced by each independent source acting alone, with all other independent sources replaced by their internal impedances.

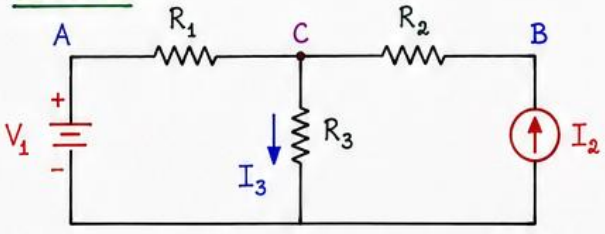
2. Applicable Conditions :

- The network must be linear.
- The sources must be independent.
- It is applicable for voltage and current calculations.

3. How to Apply :

① Consider one independent source.
② Replace all other independent voltage sources by short circuits.
③ Replace all other independent current sources by open circuits.
④ Find the required current or voltage.
⑤ Repeat the process for each independent source.
⑥ Algebraically sum all the individual contributions.

4. Example :



Find current I_3 using superposition theorem.
Given : V_1 , I_2 , R_1 , R_2 , R_3

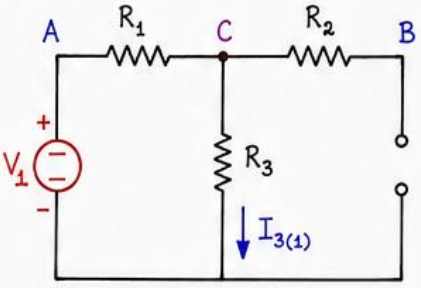
5. Derivation :

Let the total current through R_3 be I_3.
Let $I_{3(1)}$ be the current due to V_1 acting alone and $I_{3(2)}$ be the current due to I_2 acting alone.

$$I_3 = I_{3(1)} + I_{3(2)}$$

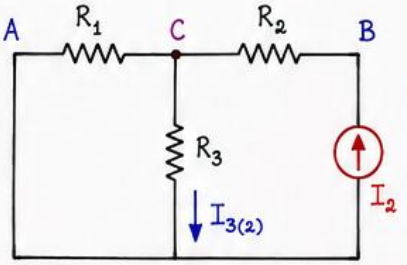
(By Superposition Theorem)

(i) Effect of V_1 acting alone (I_2 opened)



$$I_{3(1)} = \frac{V_1 R_2}{R_1R_2 + R_2R_3 + R_3R_1}$$

(ii) Effect of I_2 acting alone (V_1 shorted)



$$I_{3(2)} = \frac{I_2 R_1}{R_1R_2 + R_2R_3 + R_3R_1}$$

(iii) Total current through R_3

$$I_3 = I_{3(1)} + I_{3(2)}$$

$$= \frac{V_1 R_2}{R_1R_2 + R_2R_3 + R_3R_1} + \frac{I_2 R_1}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$= \frac{V_1 R_2 + I_2 R_1}{R_1R_2 + R_2R_3 + R_3R_1}$$

Thus, current through R_3 is the algebraic sum of currents produced by each independent source acting alone.

Notes :

- For deactivating : Voltage source → Short circuit
  Current source → Open circuit
- The sign of each term depends on the assumed direction.
- Superposition theorem cannot be applied directly to power.

6. Summary :

★ Deactivate all other independent sources.
★ Find the response due to one source.
★ Repeat for all sources.
★ Add all individual responses algebraically.
★ Applicable only to linear networks.

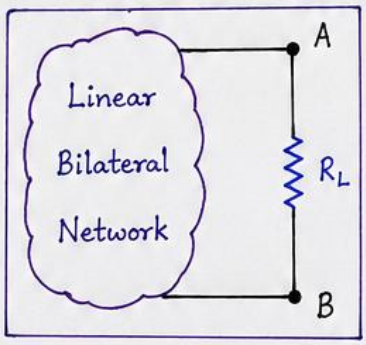
Thevenin's Theorem

Thevenin's theorem states that any linear, bilateral two terminal containing independent and dependent sources and resistances can be replaced by an equivalent circuit consisting of a single voltage source V_{th} in series with a resistance R_{th}.

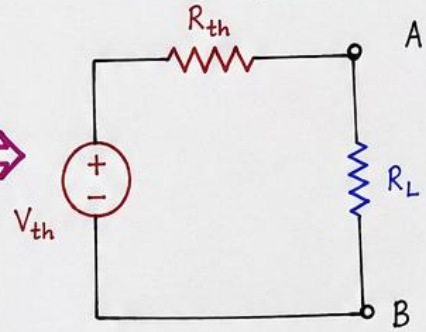
Statement :

Any linear network seen from two terminals A and B can be replaced by an equivalent circuit consisting of a single voltage source V_{th} in series with a resistance R_{th}.

Original Network



Thevenin Equivalent



How to find Thevenin Equivalent :

1. Remove the load resistance R_L from the network.
2. Find the open circuit voltage across the terminals A and B.

   $\boxed{V_{th} = V_{oc}}$ (voltage across A–B with open circuit)
3. Deactivate all independent sources (voltage sources → short circuit, current sources → open circuit). Keep dependent sources as it is.
4. Find the equivalent resistance seen from terminals A and B.

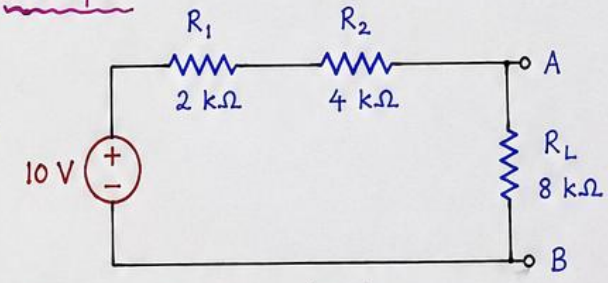
   $\boxed{R_{th} = \text{Resistance seen into A–B}}$
5. Replace the original network by V_{th} in series with R_{th} and reconnect the load R_L.

Important Points :

- V_{th} is the open circuit voltage at the terminals.
- R_{th} is the resistance seen into the network with all independent sources deactivated.
- The theorem is applicable to both DC and AC circuits.
- It is very useful for analyzing circuits with different values of load resistance.

Thus, the network is equivalent to a 6.67 V source in series with 1.33 kΩ resistance.

Example :



Open circuit voltage (V_{th}) :

$$V_{th} = 10 \times \left(\frac{R_2}{R_1 + R_2}\right) = 10 \times \left(\frac{4}{2+4}\right)$$

$$\boxed{V_{th} = 6.67\ V}$$

Thevenin resistance (R_{th}) :

$$R_{th} = R_1 \parallel R_2 = \frac{2 \times 4}{2+4} = \frac{8}{6} = 1.33\ k\Omega$$

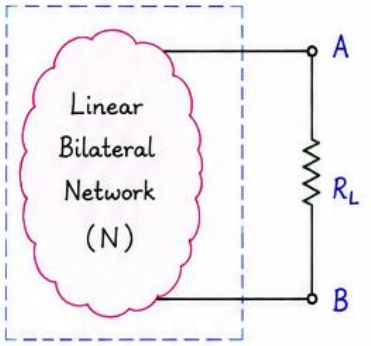
$$\boxed{R_{th} = 1.33\ k\Omega}$$

Thevenin's Theorem and Derivation

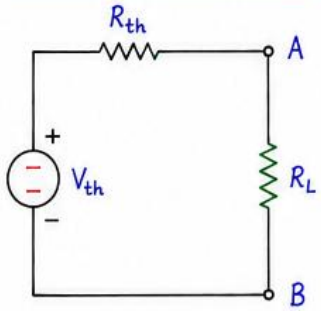
1. Thevenin's Theorem :

Thevenin's theorem states that any linear bilateral two-terminal network containing independent sources and resistances can be replaced by an equivalent circuit consisting of a single voltage source V_{th} in series with a single resistance R_{th} as shown in the figure.

Original Network



Thevenin Equivalent Circuit



V_{th} = Thevenin voltage (open circuit voltage)
R_{th} = Thevenin resistance

2. Procedure to find Thevenin Equivalent :

① Remove the load R_L from the network and find the open circuit voltage across the terminals A−B. This is V_{th}.

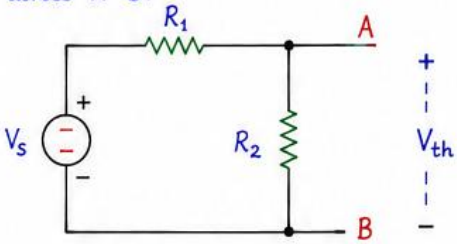
② Deactivate all independent sources in the network. (Voltage sources → short circuit, Current sources → open circuit)

③ Find the equivalent resistance seen from terminals A−B. This is R_{th}.

④ Replace the network by V_{th} in series with R_{th} and reconnect the load R_L.

3. Finding V_{th} (Open Circuit Voltage) :

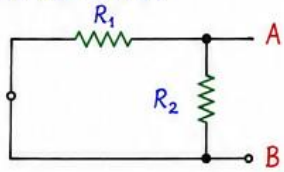
Remove R_L and find the open circuit voltage across A−B.



$$V_{th} = V_S \times \frac{R_2}{R_1 + R_2}$$

4. Finding R_{th} :

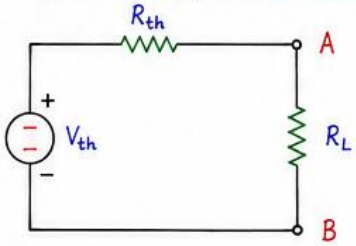
Deactivate the independent sources (V_S → short circuit) and find the equivalent resistance seen from terminals A−B.



$$R_{th} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

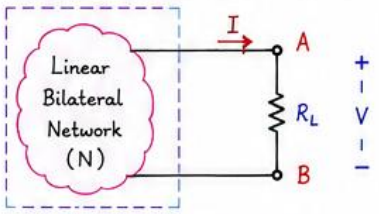
($R_1 \parallel R_2$ = parallel combination)

5. Thevenin Equivalent Circuit :



6. Derivation of Thevenin's Theorem :

Consider a linear bilateral network N with terminals A−B and a load R_L connected across it. Let the terminal voltage be V and the load current be I.



By KVL,

$$V = E - I R_{th} \quad \text{---- (1)}$$

where, E = open circuit voltage across A−B = V_{th}

R_{th} = resistance seen from A−B with independent sources deactivated.

From (1), current through load,

$$I = \frac{V}{R_L} = \frac{E - I R_{th}}{R_L}$$

Thus, the original network N behaves exactly like a circuit consisting of a voltage source E (= V_{th}) in series with a resistance R_{th}.

Hence Thevenin's theorem is proved.

Since $I = \frac{V}{R_L}$.

Substitute in (1)

$$V = E - \left(\frac{V}{R_L}\right) R_{th}$$

$$V\left(1 + \frac{R_{th}}{R_L}\right) = E$$

$$V = E \times \frac{R_L}{R_L + R_{th}} \quad \text{---- (2)}$$

★ This theorem is very useful for simplifying complex circuits and analyzing the effect of varying loads.

7. Steps Summary :

Remove R_L and find open circuit voltage V_{th}. → Deactivate all independent sources and find R_{th} seen from A−B. → Replace the network by V_{th} in series with R_{th}. → Reconnect the load R_L to get the Thevenin equivalent circuit.

8. Special Cases :

★ If no independent source, then $V_{th} = 0$.

★ If dependent sources are present, keep them active while finding R_{th}.